# Bridging Gender Disparities in Mental Health: A Bilingual Large Language Model for Multilingual Therapeutic Chatbots

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***Abstract*—Significant gender disparities and a general lack of support systems compound the state of mental health in developing countries. Women, in particular, face unique challenges due to societal norms and expectations that often restrict their access to mental health services. Additionally, the cultural stigma associated with mental health issues affects both men and women but can be particularly debilitating for women who may be less likely to seek help or disclose their struggles. This study explores the application of Jais-13B, a pre-existing bilingual Large Language Model (LLM) proficient in English and Arabic, tailored to address mental health inquiries. We employed a rich dataset encompassing both languages, essential for fine-tuning and enhancing Jais-13B's responsiveness and empathetic engagement in mental health contexts. The optimization focused on improving the model's ability to deliver accurate and contextually aware responses, which is crucial for effective mental health support. Performance evaluation using the cross-entropy loss function demonstrated that Jais-13B achieved an accuracy of 70.58%, indicating its capability to produce relevant and precise answers to mental health questions. These results showcase Jais-13B's potential to enhance digital mental health interventions significantly, providing accessible care solutions across diverse linguistic landscapes. This study reaffirms the effectiveness of bilingual LLMs in mental health applications and provides insights into their practical implementation, setting a foundational benchmark for future research in multilingual therapeutic chatbots.**

***Index Terms*—Large Language Models, Bilingual LLM, Women's Mental Health Care, Arabic Language Processing, Mental Health Chatbots.**

## I. Introduction

Witnessing the emergence of Large Language Models (LLMs), such as GPT-4 [1] and ChatGPT [2], has been a remarkable experience. These powerful virtual entities are transforming various sectors, including the mental health industry, through their exceptional ability to process and understand vast amounts of information. It is impressive how these models can execute various tasks—from language translation to solving complex logical problems—without specific training. However, a compelling question remains: Can these models truly grasp human emotions, especially those related to mental health challenges?

Although mental health issues affect many lives, they are still burdened by stigma and complexity. Anxiety and depression are not merely transient feelings; they are significant health conditions that impact millions of people worldwide. Depression is one of the most prevalent mental illnesses, affecting approximately 1 billion individuals globally, according to the World Health Organization. This widespread prevalence profoundly impacts the quality of life for many and substantially strains healthcare systems worldwide.

Amid these challenges, Large Language Models (LLMs) are gaining attention for their potential in deciphering the nuanced language of mental health. Researchers have been rigorously testing these models, evaluating their ability to understand texts related to stress, depression, and even suicidal thoughts. The results are promising, showing that LLMs can detect subtle cues indicative of mental health conditions.

Despite these challenges, LLMs are becoming increasingly prominent and have shown significant capability in understanding the complex language associated with mental health. These models have been rigorously tested by researchers who are assessing their proficiency in interpreting writings about stress, depression, and even suicidal thoughts. The encouraging results indicate that LLMs are capable of detecting the subtle signs that signify mental health issues.

Egypt is undergoing a severe crisis in mental health treatment, marked by high prevalence rates, significant social stigma, a dire shortage of mental health professionals, and uneven distribution of these specialists. This crisis is compounded by the fact that about one in four Egyptians may suffer from mental health comorbidities, such as anxiety, mood disorders, and substance use issues. Yet, many opt for traditional healers over professional psychiatric services due to factors like accessibility, affordability, and societal acceptance, further exacerbated by the stigma associated with mental illness that inhibits many from seeking formal treatment [3].

In this research, we employed LLM techniques to assist in diagnosing mental health issues in patients. We utilized

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Arabic datasets that were carefully scraped and compiled to ensure linguistic and cultural relevance. These datasets were then used to train and refine the Jais-13B model [4], resulting in a robust LLM. This model aids patients and healthcare providers by enhancing the understanding of mental health issues. This approach demonstrates how LLMs can transform mental health care by improving diagnostic processes and ensuring the accurate recognition and treatment of complex mental health conditions.

## II. Related Work

Research on LLMs in mental health care has seen a significant increase over time, primarily categorized into three areas: *Prompt-tuning and Application*, *Model Development and Fine-tuning*, and *Dataset and Benchmark*. The related work is organized into two subsections: datasets used for fine-tuning and fine-tuned models.

### A. Datasets

LLMs are extensively studied for their potential in mental health care, addressing a wide spectrum of mental health issues and providing contextually aware responses. Applications include supporting individuals on specific platforms, relationship therapy, and aiding healthcare professionals with detailed analyses and recommendations [5]–[8]. Common research topics include stress, depression, suicide prevention, anxiety, bipolar disorder, and Autism Spectrum Disorder (ASD). Life situations like relationship challenges and bullying are also studied [5], [6].

Datasets are diverse, focusing on tasks such as mental health condition detection, stress reason identification, and empathetic dialogue generation. Sources include social media platforms (e.g., Reddit, Twitter) and controlled environments (e.g., counseling sessions) [9]–[12]. Human evaluations involve experts, laypersons, or crowd workers [14], [17], [24]. While most datasets are publicly available, some are proprietary with licensing restrictions [7], [25]–[28].

### B. Fine-Tuned Models on Datasets

Recent research explores using GPT-3 [29] and GPT-4 [1] for mental health applications like depression detection, suicide prevention, and relationship counseling. Techniques such as few-shot prompting, chain-of-thought (CoT) prompting, and novel methods like chain-of-empathy prompting have been introduced to enhance performance in mental health tasks.

Fine-tuning existing LLMs on mental health-specific texts is a key focus. Examples include MentaLLaMA and Mental-LLM, which adapted LLaMA-2 [30] and Alpaca 79/FLAN-T5 models, respectively, using social media data. ChatCounselor and ExTES-LLaMA fine-tuned the LLaMA model using specialized datasets like Psych8k for improved domain-specific predictions. All fine-tuning studies employed Instruction Following Tuning (IFT), enabling models to follow instructions while incorporating domain expertise.

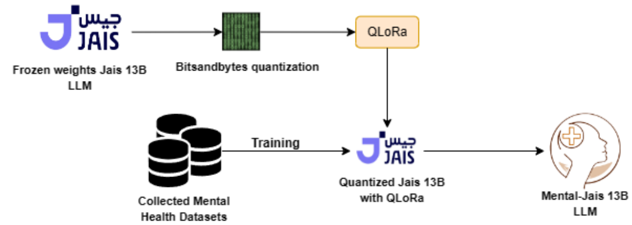


Fig. 1. Overview of the proposed LLM training pipeline.

## III. Materials and Methods

The methodology adopted in this study is illustrated in Figure 1. We selected Jais-13B as the foundational Large Language Model (LLM) [31] for our project. Jais-13B is an advanced, open-source LLM comprising 13 billion parameters designed to set benchmarks in Arabic natural language processing (NLP). This model demonstrates exceptional proficiency in processing Arabic text, displaying a deep understanding and generation capabilities while performing robustly in English-language tasks. Initially, the model underwent quantization using bitsandbytes [32] to optimize computational efficiency and reduce memory footprint in the virtual random-access memory (vRAM), facilitating operation on systems with modest computational resources.

Furthermore, the model is integrated with QLoRA [33], which plays a pivotal role in freezing the pre-trained weights of the LLM. Additionally, it strategically analyzes and modifies only a tiny practical portion of injected trainable low-rank matrices. This approach leverages the pre-existing weights effectively during the training phase with our curated datasets.

The collected datasets encompass a diverse range of sources, including pre-existing datasets and data meticulously extracted from multiple online resources in both Arabic and English, with the English portions subsequently translated into Arabic.

Finally, upon completing the training phase, the resulting model emerges as Mental-Jais 13B, an optimized domain-specific LLM specifically tailored for mental health-related conversations.

### A. Dataset Collection

Our data collection and preparation approach was carefully designed to ensure that the model performed well in predicting the data used. This included a solid effort to specialize in Arabic language facts, mainly focusing on question-and-answer pairs such as patient-to-doctor communication. This was important for our research in mental health analysis because novelty and open-ended questions were scarce among our data preferences. We relied heavily on publicly available datasets, combining web-scraping and translation techniques.

The primary mission focused on obtaining Arabic content, thus requiring a significant web-scraping effort. We have carefully curated information from various mental health websites, each of which is a rich source of information about psychological disorders. Notably, we gleaned valuable insights from the Substance Abuse and Mental Health Services Administration

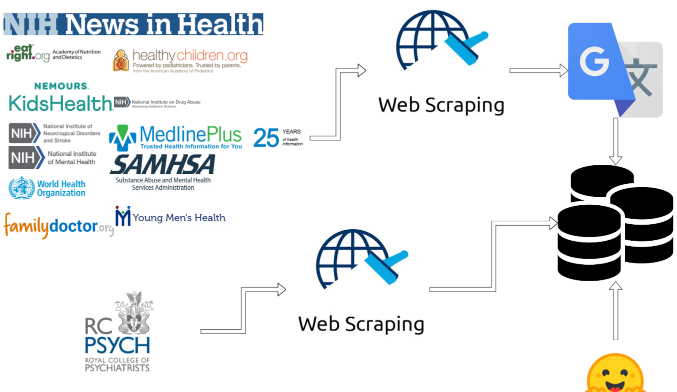


Fig. 2. Dataset collection pipeline.

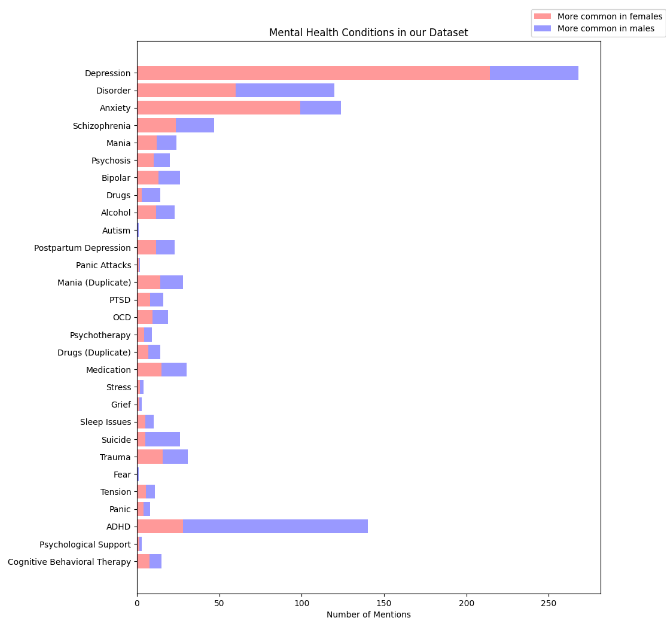


Fig. 3. Scraped diseases.

(SAMHSA) website [34], which provided comprehensive information on a spectrum of mental health conditions including but not limited to Antisocial Personality Disorder, Anxiety Disorders, Bipolar Disorder, Depression, and Schizophrenia.

We obtained data from reputable international organizations such as the World Health Organization (WHO) [35] and MedlinePlus [36], enriching our dataset with common mental disorders such as anxiety, depression, and eating disorders. In addition, we searched specialized resources such as KidsHealth [37] and the National Institute of Mental Health (NIMH) [38] for insights tailored to specific populations, including adolescents, from ADHD to body image problems.

To ensure comprehensiveness, we also included data from federal health agencies such as the National Institutes of Health (NIH) [39] and reputable medical societies such as the American Academy of Pediatrics [40]. These sources furnished us with insights into various mental health issues and treatments, including ADHD, depression, and positive body image promotion for children. Moreover, we used translations from authoritative sources such as the Royal College of Psychiatrists [41], which provide valuable information on mental health in Arabic, including depression and anxiety, for our dataset.

In addition to the data mentioned earlier, we gathered insights from reputable health news organizations such as NIH's News in Health [42], which expanded our dataset to include issues such as parenting, peer relationships, and body odor. Furthermore, our English downloads were translated into Arabic using the Google translation tool [43]. We found Google Translate to be more accurate in translation than other tools, including ChatGPT translation tools, hence the preference in our data translation process. We supplemented our dataset with data from Hugging Face, containing mental health conversational datasets [44]–[46], which were translated into Arabic using similar translation techniques. It is worth noting that each data point collected underwent rigorous pre-processing, ensuring linguistic coherence and consistency across the dataset.

In summary, our data collection approach involved a meticulous curation process drawing from a diverse array of reputable sources spanning governmental agencies, international organizations, specialized medical associations, and health news outlets, as shown in Figure 2, all aimed at creating a robust and comprehensive dataset for our research, as illustrated in the graph of Figure 3.

### B. Proposed Model Architecture for Jais-13B

Jais-13B is a comprehensive language model developed through a collaborative effort between G42's Inception Institute, Mohamed bin Zayed University of Artificial Intelligence, and Cerebras Systems. Conceived upon the foundation of the GPT-3 [29] architecture, Jais-13B exhibits exceptional proficiency in Arabic and English. It demonstrates a profound understanding and adept generation of Arabic text while maintaining robust capabilities in various English-based tasks. The model was meticulously trained on a vast Arabic and English text corpus. Subsequently, a specialized version, designated as Jais-13B chat, was fine-tuned to facilitate the execution of instructions and engage in conversational interactions. This enhanced version is predominantly employed for conversational purposes, and it is the one we used to create Mental-Jais-13B.

The Jais-13B LLM consists of 40 blocks known as "Jais Blocks." Each block comprises multiple sub-components that process and transform the input data through various stages. The proposed architecture represents blocks that have several other small components, which, added together, enhance model performance and stability: normalizing inputs via Layer Normalization, JAIS Attention deals with self-attention and incorporates mechanisms for dropout and projection for greater focus without overfitting; a multi-layer perceptron module that processes the data through multiple transformations, with enhanced non-linearity given through SwiGLU and further

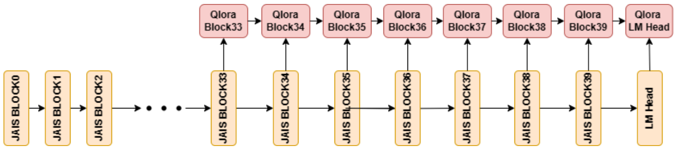


Fig. 4. Proposed model architecture with quantization and QLoRA.

leveraging dropout to prevent overfitting. On top of that, alibi position embeddings capture the token order, while the final LM head projects the processed data into the vocabulary space to give logits for the language modeling task.

Each Jais Block integrates these components to enable the model to perform complex transformations and learning tasks on the input data, exploiting attention mechanisms and feed-forward neural networks.

Our proposed study presents an architecture that utilizes quantization and QLoRA, as described in Figure 4. The main contribution lies in strategically selecting target modules for the QLoRA application. This selection is crucial in environments with modest computational power, where creating an entirely new version of the quantized model using QLoRA is challenging. Therefore, we chose the last seven Jais blocks and the LM HEAD block as the QLoRA target modules, which was intuitive as it follows the idea of ”adding a small model on top of the foundation model” to ensure smooth fine-tuning. This approach allowed the study to be conducted using approximately 20 GB of VRAM, as opposed to the more than 100 GB of VRAM required for float/bfloat 16-bit precision or 200 GB of VRAM in the case of float32 precision.

The hyperparameters used for Jais-13B chat that can be found in our Hugging Face repository [47], [48].This hyperparameter choice was based on multiple experiments that prioritized performance for modest computational power, especially for the selection of the `max_new_tokens` parameter while also maintaining the stability of the model in answering questions.

### C. Quantization Using Bitsandbytes

Quantization is a formal technique employed to compress models by representing the weights and activations of the neural networks in the form of lower-precision numerical values. LLMs generally utilize 32-bit floating-point numbers (FP32) for their weights, and the quantization process is responsible for compressing these weights into formats with lower precision, frequently 8-bit integers (INT8) or even 4-bit integers (INT4), the latter of which we utilized. The ”bitsandbytes” package was employed in the proposed approach, which handles the quantization process stably and consistently. This is accomplished through the following steps:

1) **Scaling**: Each weight tensor, a multi-dimensional array of numbers, is analyzed to determine its maximum absolute value. This value is then used as a scaling factor to adjust the weights.
2) **Quantization**: The weights are divided by the scaling factor and rounded to the nearest integer within the representable range of the target data type (e.g., -128 to 127 for INT8).
3) **Storage**: The quantized integers and their respective scaling factors are stored, replacing the original FP32 weights.

This process significantly aids in training large language models on constrained resources while preserving the model's performance by minimizing the memory footprint of the graphics card's VRAM.

Furthermore, it is imperative to acknowledge that a process known as dequantization takes place during the model inference. While the model is stored using the quantized format, calculations are strategically dequantized (converted back to higher precision) during inference. This allows the model to uphold a satisfactory level of accuracy notwithstanding compression.

### D. PEFT and QLoRA

Parameter-efficient fine-tuning (PEFT) encompasses a variety of techniques aimed at fine-tuning large language models with reduced computational costs. It focuses on minimizing the number of trainable parameters to make the process more efficient. Quantized Low-Rank Adaptation (QLoRA) is a cutting-edge type of PEFT designed to fine-tune massive LLMs efficiently through LoRA principles (Low-Rank Adaptation).

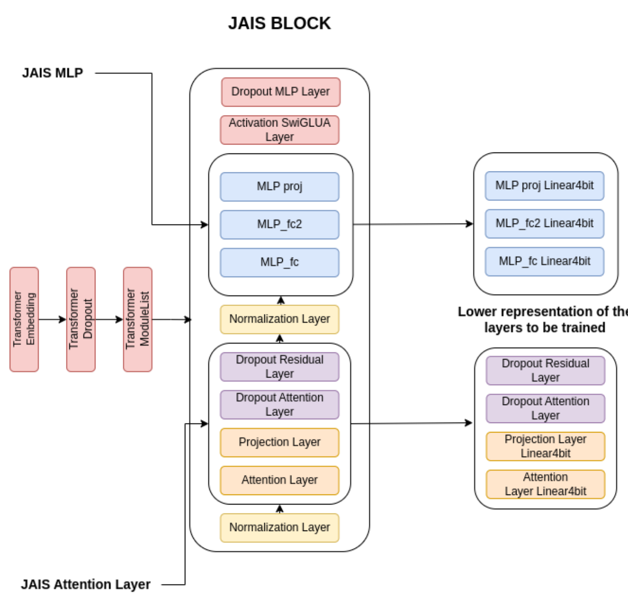


Fig. 5. QLoRA demonstration figure.

As illustrated in Figure 5, LoRA injects trainable, low-rank decomposition matrices into each LLM transformer architecture layer. This technique focuses updates on a reduced set of parameters instead of modifying the entire model, significantly decreasing the number of trainable parameters. Building upon this, QLoRA further enhances efficiency by quantizing the LLM's weights. It employs ”NormalFloat4” (NF4), a 4-bit quantization scheme, compressing weights from their usual 32-bit representation. QLoRA employs double quantization for additional storage savings, even quantizing the quantization constants. To mitigate potential memory strains during training, QLoRA integrates paged optimizers for streamlined memory management.

It is important to note that the computation of the original neural network's complete forward pass remains necessary. We can conserve resources during the backward pass (backpropagation step) because only the newly trainable LoRA layers require gradient updates.

The QLoRA used parameters are illustrated in our Hugging Face repository [47], [48].. It's important to note that the rank of the LoRA matrices determines the dimensionality of the added matrices $A$ and $B$ in each layer of the transformer. A higher rank $r$ generally allows for capturing more complex adaptations and increases the number of trainable parameters. The rank $r$ defines the space size for LoRA modifications.

Alpha (scaling factor): The LoRA matrices modify the existing calculations within the LLM, and the alpha acts as a volume knob, determining how strongly those modifications influence the final output of the model. In other words, the magnitude of change can be considered, where the alpha parameter determines how dramatically the LoRA matrices are allowed to reshape the model's output within that adaptation space.

Lastly, the `lora_dropout` parameter ensures that the LLM is not dependent on the LoRA adaptations, encouraging it to update the underlying model weights for better generalization.

In this proposed study, we created QLoRA blocks, which have the same components as the JAIS blocks but are quantized to a precision level of 4 bits. As previously mentioned, we selected the last seven JAIS blocks as our QLoRA target modules. This selection effectively fine-tuned the model to our created dataset using modest computational power, achieving impressive results.

## IV. Results

Considering the amalgamation of utilized technologies, the Mental-Jais-13B model was presented. It had undergone training on our collected dataset designed for engaging in dialogue with users suffering from mental health issues in Arabic. This language model exhibits proficiency in conversing with users and endeavors to address their concerns by providing guidance and strategies for managing their illnesses based on a knowledge base that our dataset trained on.

As demonstrated in the figures available on our Hugging Face repository [47], [48]., the proposed Mental-Jais model was able to answer questions more effectively than the vanilla Jais model. The answers were based on the knowledge base of the created dataset, allowing the model to respond accurately. It is worth noting that both models were given the same prompt, which explicitly instructed them that it was acceptable to provide mental health advice while recommending seeking professional help.

For instance, the vanilla Jais model refused to answer in the second record, likely because it lacked relevant information. In contrast, the proposed model responded professionally based on its knowledge base. In the third user prompt, both models followed a similar approach. However, in the fourth user prompt, the vanilla Jais model could not answer the question due to a lack of knowledge, while the proposed model could respond effectively based on its knowledge base derived from the collected dataset.

In assessing the performance of the Mental-Jais-13B model, it was able to score 30.42 loss; since the used loss function is the cross-entropy loss, it is safe to say that the model scored more than 70.58% accuracy on the collected dataset. This accuracy demonstrates the amount of information captured by the model of the dataset. While this might indicate a moderate level of performance, this is highly affected by the accuracy of the Arabic language and its nature of scoring relatively low accuracy values during the training due to the difficulty and richness of the Arabic language.

## V. Discussion

The findings of our study highlight the use of the Arabic LLM Jais-13B, which offers bilingual support in mental health. The proposed Mental-Jais-13B specializes in answering questions related to mental health illnesses and helping users based on the dataset it was trained on. The model achieved an accuracy rate of 70.58%, as the cross-entropy loss function indicated. This level of accuracy is notable given the inherent complexities and nuances of the Arabic language, which is known for its rich morphological structure and contextual depth. The model's performance suggests it can effectively interpret and respond to mental health queries, providing empathetic and contextually appropriate responses. LLMs in mental health care can be particularly beneficial in regions where such services are scarce or stigmatized.

## VI. Conclusion

This study has robustly demonstrated that bilingual large language models proficient in both English and Arabic are formidable assets in digital mental health interventions. Achieving an accuracy of 70.58%, Jais-13B has proven its capacity to deliver empathetic and precise responses and culturally attuned feedback to a diverse range of mental health inquiries. The model's dual-language capabilities enhance accessibility for a wider audience, thereby making mental health services more inclusive and reducing linguistic barriers that have traditionally hindered care provision.

Further integration of sophisticated NLP capabilities, such as advanced sentiment analysis and contextual understanding, could refine the model's ability to discern and adapt to users' emotional states, thereby enhancing the quality of interactions. Practical field testing in clinical and informal settings will be essential to evaluate Jais-13B's real-world effectiveness and gather user feedback for continuous improvement.

Collaborating with mental health professionals to adapt model responses to align more closely with clinical guidelines could ensure that the chatbot's advice supports therapeutic outcomes. Addressing ethical considerations, particularly data privacy and the ethical use of AI in sensitive areas like mental health, will remain a priority in building trust and ensuring the responsible deployment of these technologies.

## Data & Code Availability

The collected dataset and the QLoRA adapters and weights will be available on Hugging Face [47], [48].


## Acknowledgment

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